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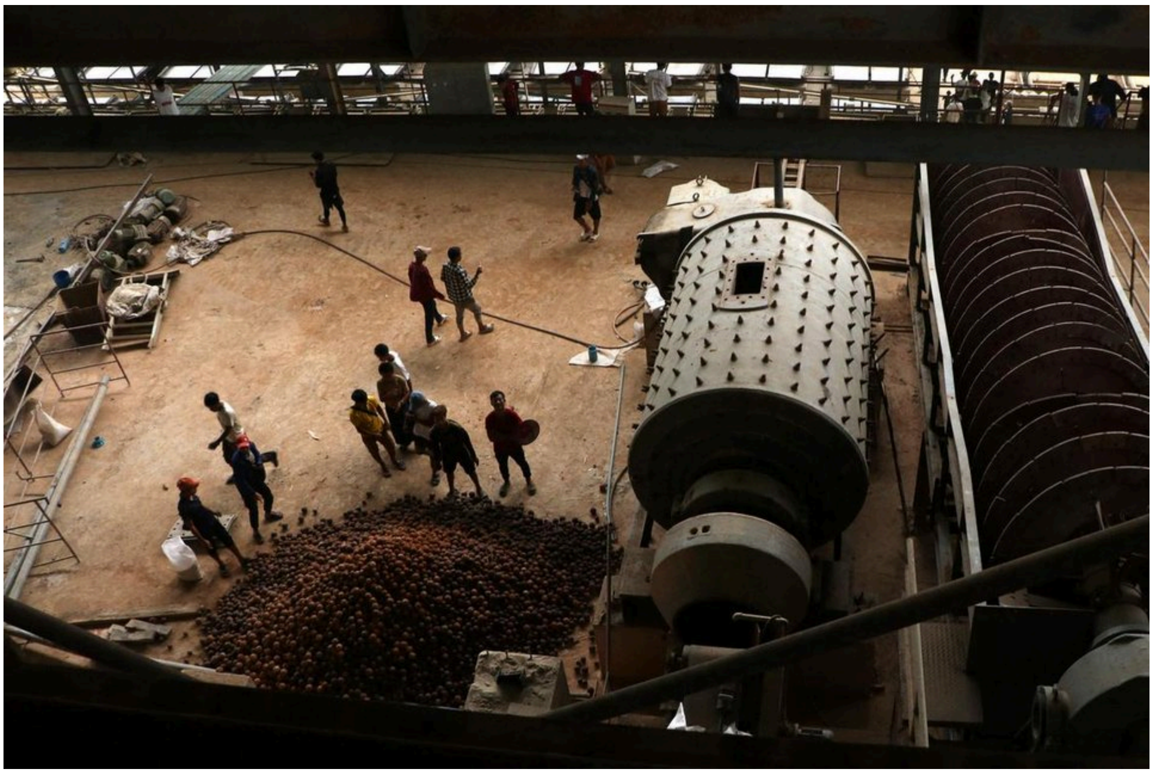
China's green paradox: Cleaner at home, digging dirt abroad

Beijing's rush to secure minerals from poor countries renews questions on whether it can truly be a global climate leader.



Members of the Pradawng tribe inside a China-backed battery metal mine during a protest rally in Pekon township in Myanmar's eastern Shan State in May.

PHOTO: AFP

Tan Dawn Wei

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Published Aug 07, 2025, 05:00 AM
Updated Aug 07, 2025, 05:45 AM

As the world grapples with the grave effects of climate change and the urgent task of dealing with it, China has provided some optimism with signs that its carbon emissions might have peaked, even as its demand for power surges at home.

Recent analysis by the Centre for Research on Energy and Clean Air (CREA) showed that for the first time, China's carbon emissions have declined thanks to its solar, wind and nuclear energy investments, a growing share of electric vehicles on the road and a drop in cement and steel production.

While there have been previous dips, they were caused by economic slowdown and the Covid-19 pandemic. But this time around, the drop has occurred even as the country's electricity demand continues to grow – a signal that the world's largest emitter may be decoupling economic growth from carbon pollution.

Beijing has set a target for peaking carbon emissions by 2030 and achieving carbon neutrality by 2060.

Granted, the decline might be slight – emissions fell by 1 per cent over the past 12 months, including a 1.6 per cent drop in the first quarter of 2025 – and there is the possibility that it could rebound to a record high, said the report's author, CREA lead analyst Lauri Myllyvirta.

But if current trends hold, China could be reaching a turning point that would mark a symbolic and strategic breakthrough for global climate action and empower China to take on a stronger leadership role.

Yet, there is a paradox behind this promising trajectory. In July, a report released by Griffith Asia Institute and Fudan University's Green Finance and Development Centre showed that China's engagement in projects under its Belt and Road Initiative (BRI) was the highest in any six-month period, with US$66.2 billion (S$85.2 billion) in construction contracts and US$57.1 billion in investments.

Of these, its green energy engagement hit new highs with US$9.7 billion in wind, solar and waste-to-energy projects and an installed capacity of 11.9GW. But it also continued to build coal mine infrastructure, while its investments in the highly polluting metals and mining sector in the first six months of 2025 alone trumped the whole of 2024, already a record year.

As China endeavours to reach its climate goals, but also aggressively secure critical minerals and metals to fuel its industrial economy and energy transition, while reducing dependency on Western-controlled supply chains, one question has resurfaced.

Can it credibly claim to be a global leader in the fight against climate change if its decarbonisation efforts at home are effectively being outsourced to poorer, less-regulated countries.

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Green energy saviour

There's no denying the immense progress Beijing has made in greening its economy and the global contributions it has made with its clean technology.

Since 2021, China has added more solar capacity annually than the rest of the world combined. In April 2025, its solar and wind power generation capacity exceeded thermal power capacity for the first time in its history, its energy regulator said.

A separate report by CREA's Mr Myllyvirta published recently showed that China's exports of clean tech like solar panels, electric vehicles and batteries in 2024 helped shave off 1 per cent in global emissions outside of China.

While its clean energy footprint spans the world, its exports and investments are benefiting regions like sub-Saharan Africa, where emissions could be cut by 3 per cent, and in the Middle East and North Africa, which could have a 4.5 per cent reduction.

Its exports – US$177 billion in solar panels, EVs, batteries and wind turbines in 2024 – could continue to increase if it can hold on to its market share, said Mr Myllyvirta.

Even so, there is a caveat to the notion that Chinese green exports are helping to save the world from the effects of global warming.

"An interesting question is whether the underlying reasons are about climate change or owning the economy of the future, which because of poor policy choices in the US and Europe, could leave other countries dependent on Chinese-made technology for decades to come," Mr Brad Adams, executive director of green NGO, Climate Rights International, told The Straits Times.

The quest for critical minerals

At the same time, China's strategic interest in metals and critical minerals like copper, aluminium, iron, ores, cobalt, nickel, lithium and rare earth elements is driving record investments in mining and processing from Africa and Latin America to South-east Asia.

It is the top importer of cobalt, nickel, copper and bauxite, with supply chains deeply tied to BRI-backed countries.

Nearly US$25 billion has been poured into the metals and mining sector in the first half of 2025 alone, more than any other entire year since the BRI was launched in 2013. Of this amount, Kazakhstan took the monster share, with US$12 billion going into aluminium and US$7.5 billion in copper mining and smelting.

Mining and minerals processing now account for 20 per cent of all Chinese BRI engagement, solidifying the sector as China's second-largest overseas focus after energy, the BRI report showed.

The rush to capture these vital commodities – essential in clean tech, advanced manufacturing and defence systems – reflects a deepening strategy by Beijing to build geopolitical and economic leverage, as it seeks to maintain leadership in green tech, build energy security and economic resilience.

This strategy of securing upstream resources while exporting clean tech downstream may make strategic and economic sense for China. It ensures supply chain dominance in the industries of the future, and it helps Chinese firms climb the value chain in green manufacturing.

But it also creates something of a contradiction: It is reducing its environmental footprint domestically while expanding it internationally, which raises questions about responsibility, equity, and whether it is following in the footsteps of Western colonial powers in extracting resources overseas and engaging in "environmental imperialism".

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Ecological damage

Critics have long taken aim at the BRI for relocating polluting industries to poorer countries and wreaking environmental damage with its infrastructure projects, prompting China to propose in 2017 a "Green BRI" that would promote sustainable projects.

In 2021, President Xi Jinping took this further by pledging that China will no longer build coal-fired power plants abroad.

Yet, it has continued to invest in coal activities overseas, research has shown. The recent BRI report identified over US$1.58 billion in contracts for coal-related activities. State-owned power construction company, PowerChina, is developing a coal mine in Mongolia, while China Railway is building a rail network in Indonesia to transport coal.

“The mining projects have a long way to go to live up to the requirement from China’s leadership that the BRI should be green,” Mr Myllyvirta, who has been tracking coal plants linked to metals processing, and weak emission controls in the plants and smelters, told The Straits Times.

“These are symptoms of a broader weakness in environmental governance of overseas projects, especially in countries such as Indonesia and Myanmar where environmental regulation is weak and poorly enforced.”

Imported iron ore, bauxite and coal stored at Yantai Port, in China's eastern Shandong province, on July 25 before being transported.

PHOTO: AFP

These Chinese-linked mining projects – through a mix of acquisitions, grants, loans or resource-for-infrastructure deals – have been drawing growing scrutiny over their environmental impact, sparking protests, legal action and diplomatic tensions.

In the Democratic Republic of Congo, Chinese companies that control much of the mining of cobalt, which is vital for lithium batteries in electric vehicles, and copper – used in everything from smartphones to data centres – have been accused of contaminating rivers, destroying farmland and disposing hazardous waste near communities.

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Rare earth mining in Myanmar's northern Kachin State has caused severe soil erosion, toxic waste leaks and deforestation, studies have shown.

And nickel smelting and mining in Sulawesi and Halmahera Islands in Indonesia have also led to air and water pollution, high emissions from coal-powered nickel refining and deforestation. Chinese companies control three-quarters of Indonesia's nickel refining capacity, said a report by Washington-based research group, Centre for Advanced Defence Studies, earlier in 2025.

To be sure, China is far from alone in shifting its environmental burden overseas. "Green colonialism" has been practised by many wealthy nations. The US, by outsourcing manufacturing to countries like China and Mexico, and South-east Asia, has managed to maintain relatively lower domestic emissions growth despite high consumption levels.

Japan has long invested in forest resource exploitation and fossil fuels infrastructure across South-east Asia and South Asia, which has led to deforestation and pollution. It was a major funder of coal power plants abroad, especially in Indonesia, Vietnam and Bangladesh, before pledging to exit coal financing in 2020.
"Mineral production will continue to grow, with or without Chinese involvement," said Mr Putra Adhiguna, managing director of Energy Shift Institute, an energy finance think-tank.

"In the green transition, it's fairer to ask Chinese overseas investments to outperform industry benchmarks for each metal/mining sector, while also raising the standards of their activities abroad."

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CREA's Mr Myllyvirta agrees.

"It took other international financiers decades to develop decent environmental standards that act as a backstop in countries whose own regulations are weak, and China should do this as a matter of urgency – most naturally, applying the domestic emission standards and other environmental standards that are required domestically in China, when they are stronger than those in the host countries, as they often are."

At stake is reputational damage and the risk of conflicts between local communities and Chinese investors and developers, says Mr Myllyvirta.

The global race for critical minerals is fraught with trade-offs. But China's unique role as both the world's largest polluter – responsible for about a third of humanity's greenhouse gas emissions – and its largest producer of green technology gives it disproportionate influence and responsibility.

It can help "chart a third path for development", says Mr Putra. "Admittedly, Western standards may be too costly for many emerging economies and China is well-placed to offer a more practical model by aligning its investments with stronger social and environmental safeguards."

If China wants to cement its status as a true climate leader, it must, at the very least, apply the same standards of environmental governance to its overseas investments as it does to its domestic industries.

Tan Dawn Wei is a senior columnist at The Straits Times. She was the paper's China bureau chief based in Beijing from 2018 to 2025.

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